

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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Discrete Fourier Transform

- Definition of the Discrete Fourier Transform
- Properties of the Discrete Fourier Transform
- Calculating Linear Convolution using DFT
- Analyzing Signal Spectra using DFT
- Calculating DFT using MATLAB

Finite-Length Sequence Fourier Analysis

The Fourier transform theoretically addresses how to map from the time domain to the frequency domain

Continuous Aperiodic Signals $x(t)$:

$$X(\mathrm{j}\omega)=\int_{-\infty}^{+\infty}x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

Continuous Periodic Signals $\tilde{x}(t)$

$$X(n\omega_0)=\frac{1}{T}\int_{0}^{T}\tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

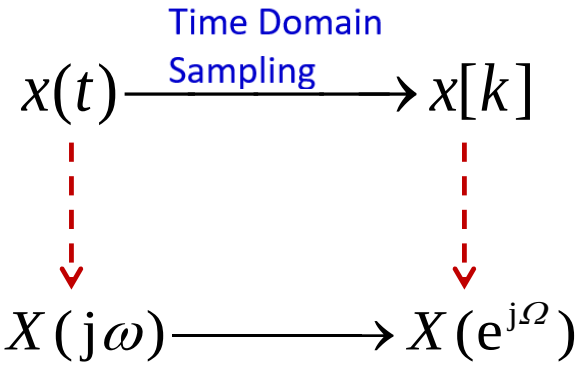
Discrete Aperiodic Signals $x[k]$:

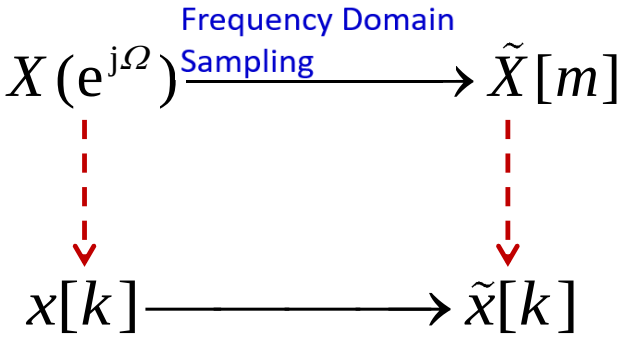
$$X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=-\infty}^{+\infty}x[k]\mathrm{e}^{-\mathrm{j}\Omega k}$$

Discrete Periodic Signals $\tilde{x}[k]$

$$\tilde{X}[m]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}$$

Finite-Length Sequence Fourier Analysis

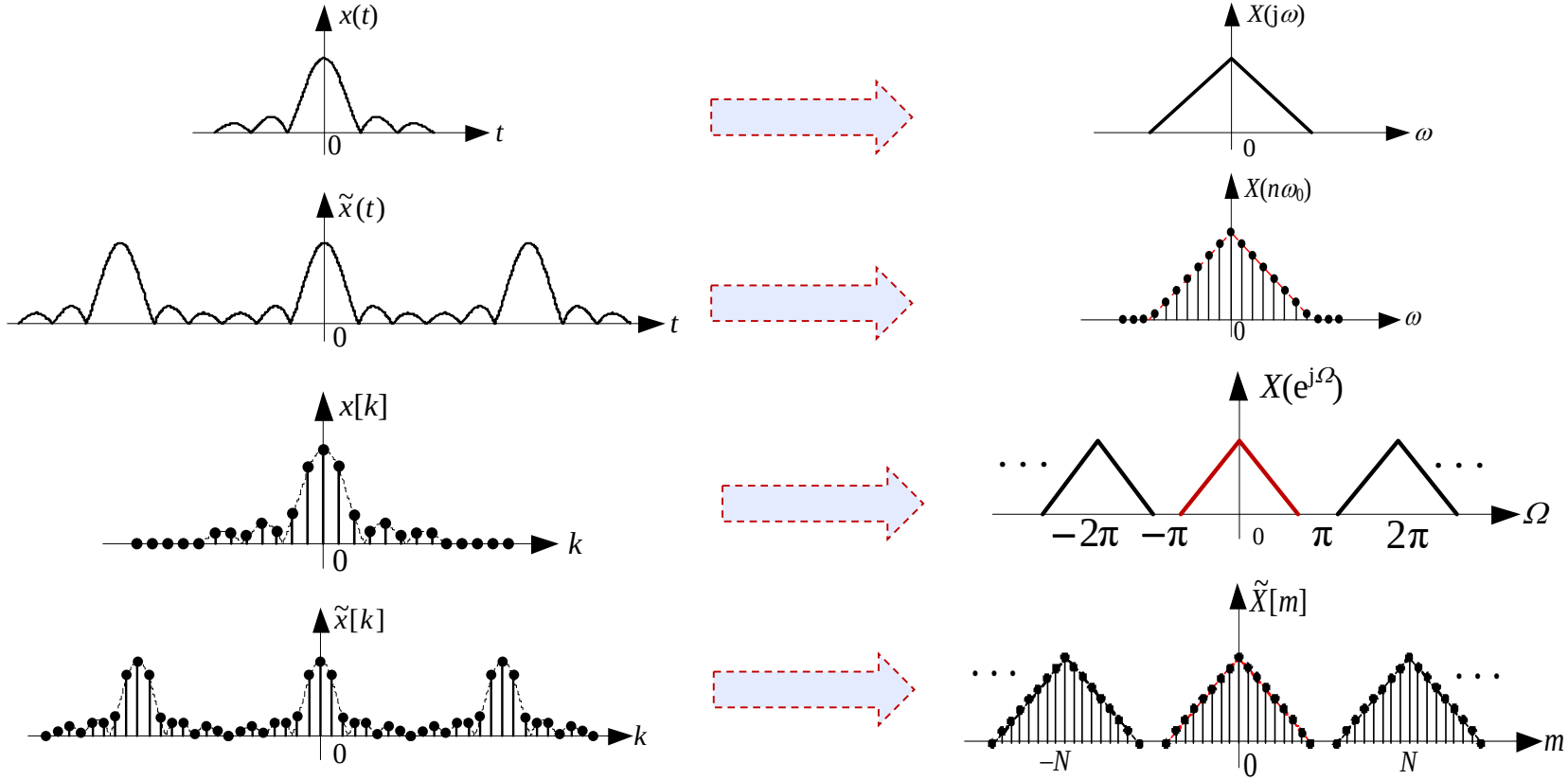




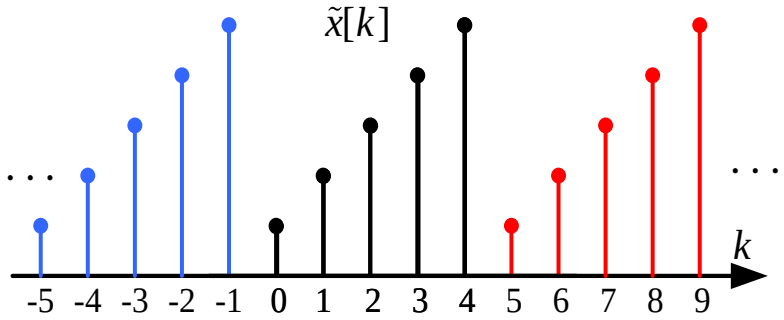
The discretization of a signal in the time domain corresponds to its periodicity in the frequency domain.

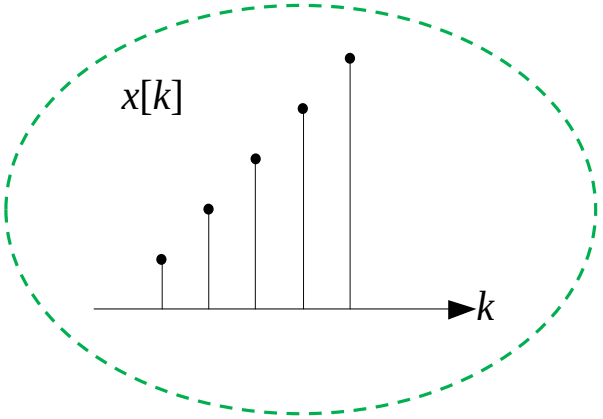
The discretization of a signal in the frequency domain corresponds to its periodicity in the time domain.

Finite-Length Sequence Fourier Analysis



Finite-Length Sequence Fourier Analysis





DFS $\tilde{x}[k] \rightarrow \tilde{X}[m] = \sum_{k=0}^{N-1} \tilde{x}[k] e^{-j\frac{2\pi}{N}mk}$

Can finite-length sequences and periodic sequences be treated equivalently in terms of analysis methods?

Definition of the Discrete Fourier Transform

DFS $$\tilde{X}[m]=\sum_{k=0}^{N-1}\tilde{x}[k]\cdot e^{-j\frac{2\pi}{N}mk}$$

DFT $$X[m]=\sum_{k=0}^{N-1}x[k]\cdot e^{-j\frac{2\pi}{N}mk}$$

$0\le m\le N-1$

Discrete Fourier Transform(DFT)

Definition of the Discrete Fourier Transform

IDFS $$\tilde{x}[k]=\frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}[m]\mathrm{e}^{\mathrm{j}\frac{2\pi}{N}mk}$$

IDFT $$x[k]=\frac{1}{N}\sum_{m=0}^{N-1}X[m]\mathrm{e}^{\mathrm{j}\frac{2\pi}{N}mk}$$ $0\le k\le N-1$

Inverse Discrete Fourier Transform

Definition of the Discrete Fourier Transform

DFT $$X[m]=\sum_{k=0}^{N-1}x[k]\cdot \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk},\qquad m=0,1,2,\cdots,N-1$$

IDFT $$x[k]=\frac{1}{N}\sum_{m=0}^{N-1}X[m]\cdot \mathrm{e}^{\mathrm{j}\frac{2\pi}{N}mk},\qquad k=0,1,2,\cdots,N-1$$

Representation:

$$X[m]=\mathrm{DFT}\{x[k]\}$$

$$x[k]=\mathrm{IDFT}\{X[m]\}$$

$$x[k]\xleftrightarrow{\mathrm{DFT}}X[m]$$

Discrete Fourier Transform

Example: Calculate the DFT of the finite-length 4-point sequence $x[k]=\{1, 1, 1, 1; k=0,1,2,3\}$.

Solution: $X[m]=\sum_{k=0}^{N-1} x[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}=\sum_{k=0}^{3} x[k]W_4^{mk}$ $W_N=\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}}$

$$X[0]=x[0]W_4^0+x[1]W_4^0+x[2]W_4^0+x[3]W_4^0=4$$

$$X[1]=x[0]W_4^0+x[1]W_4^1+x[2]W_4^2+x[3]W_4^3=0$$

$$X[2]=x[0]W_4^0+x[1]W_4^2+x[2]W_4^4+x[3]W_4^6=0$$

$$X[3]=x[0]W_4^0+x[1]W_4^3+x[2]W_4^6+x[3]W_4^9=0$$

$$X[m]=\{4,\ 0,\ 0,\ 0;\ m=0,1,2,3\}$$

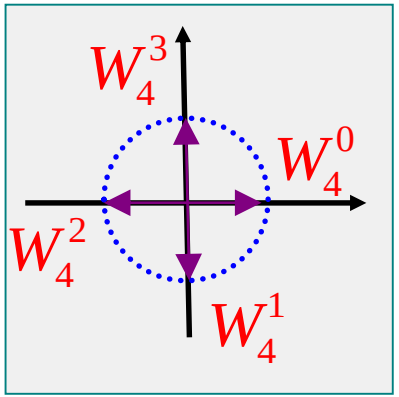
Discrete Fourier Transform

$$X[m]=\sum_{k=0}^{N-1} x[k]W_N^{mk}$$

$$W_4 = e^{-j\frac{2\pi}{4}}$$

Matrix representation of the DFT of the finite-length 4-point sequence $x[k]$

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^0 & W_4^2 \\ W_4^0 & W_4^3 & W_4^2 & W_4^1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$



$$W_4^4 = W_4^0 \quad W_4^6 = W_4^2 \quad W_4^9 = W_4^1$$

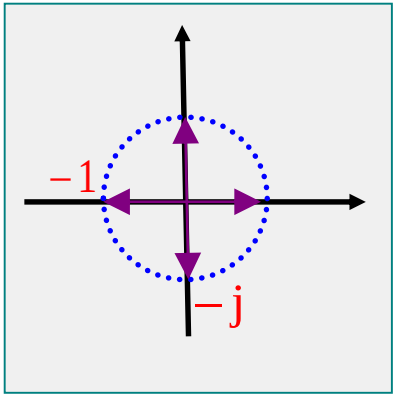
Discrete Fourier Transform

$$X[m]=\sum_{k=0}^{N-1} x[k]W_N^{mk}$$

$$W_4 = e^{-j\frac{2\pi}{4}}$$

Matrix representation of the DFT of the finite-length 4-point sequence $x[k]$

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$



Discrete Fourier Transform

Example: Calculate the DFT of the finite-length 4-point sequence $x[k]=\{1,1,1,1;k=0,1,2,3\}$.

Solution:

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$X[m]=\{4,\ 0,\ 0,\ 0;\ m=0,1,2,3\}$$

Discrete Fourier Transform

Example: Calculate the DFT of the finite-length 4-point equence $x[k]=\{1,1,1,1;k=0,1,2,3\}$.

Solution: $x[k]=\{1,1,1,1\}$ $X[m]=\{4,0,0,0\}$

What happens to the DFT of the sequence $x[k]$ when zeros are appended to make it an 8-point sequence?

$$x_0[k]=\{1,1,1,1,0,0,0,0\}$$

$$X[m]=\sum_{k=0}^{N-1} x[k]W_N^{mk}=\sum_{k=0}^{7} x[k]W_8^{mk}$$

$$X_0[m]=\{4,1-(1+\sqrt{2})\mathrm{j},0,1+(1-\sqrt{2})\mathrm{j},0,1+(-1+\sqrt{2})\mathrm{j},0,1+(1+\sqrt{2})\mathrm{j}\}$$

Discrete Fourier Transform

$$x[k]=\{1,1,1,1\},\quad 0\le k\le 3$$

$$X[m]=\{4,0,0,0\},\quad 0\le m\le 3$$

$$x_0[k]=\{1,1,1,1,0,0,0,0\},\quad 0\le k\le 7$$

$$X_0[m]=\{4,1-(1+\sqrt{2})\mathrm{j},0,1+(1-\sqrt{2})\mathrm{j},0,1+(-1+\sqrt{2})\mathrm{j},0,1+(1+\sqrt{2})\mathrm{j}\}$$

$$0\le m\le 7$$

Discrete Fourier Transform

Example: Calculate the DTFT of the sequences $x[k]=\{1,1,1,1\}$ and $x_0[k]=\{1,1,1,1,0,0,0,0\}$.

Solution: $$X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=-\infty}^{\infty}x[k]\mathrm{e}^{-\mathrm{j}\Omega k}=\sum_{k=0}^{3}x[k]\mathrm{e}^{-\mathrm{j}\Omega k}=\sum_{k=0}^{3}\mathrm{e}^{-\mathrm{j}\Omega k}$$

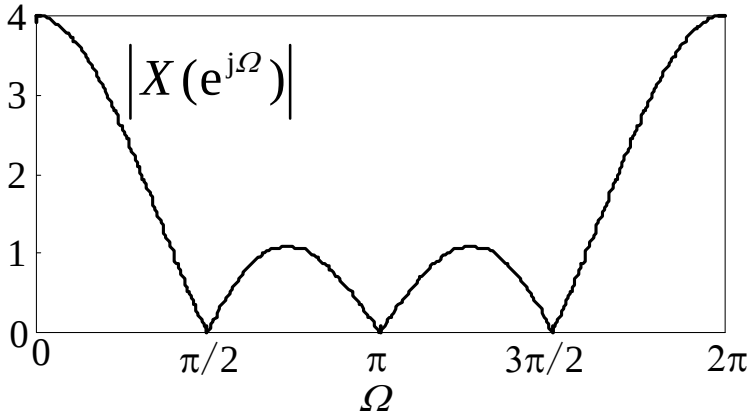
$$X_0(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=0}^{7}x[k]\mathrm{e}^{-\mathrm{j}\Omega k}=\sum_{k=0}^{3}\mathrm{e}^{-\mathrm{j}\Omega k}$$

$$X(\mathrm{e}^{\mathrm{j}\Omega})=X_0(\mathrm{e}^{\mathrm{j}\Omega})=\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}\left[2\cos(\frac{1}{2}\Omega)+2\cos(\frac{3}{2}\Omega)\right]$$

Discrete Fourier Transform

Example: Calculate the DTFT of the sequences $x[k]=\{1,1,1,1\}$ and $x_0[k]=\{1,1,1,1,0,0,0,0\}$.

Solution: $$X(\mathrm{e}^{\mathrm{j}\Omega}) = X_0(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}\left[2\cos(\frac{1}{2}\Omega)+2\cos(\frac{3}{2}\Omega)\right]$$



Discrete Fourier Transform

$$X(e^{j\Omega}) = X_0(e^{j\Omega}) = e^{-j\frac{3}{2}\Omega}\left[2\cos(\frac{1}{2}\Omega) + 2\cos(\frac{3}{2}\Omega)\right]$$

$$x[k] = \{1,1,1,1\}$$

$$X[m] = X(e^{j\Omega})\big|_{\Omega=\frac{2\pi}{4}\cdot m} \qquad \Omega = 0, \quad \frac{\pi}{2}, \quad \pi, \quad \frac{3\pi}{2}$$

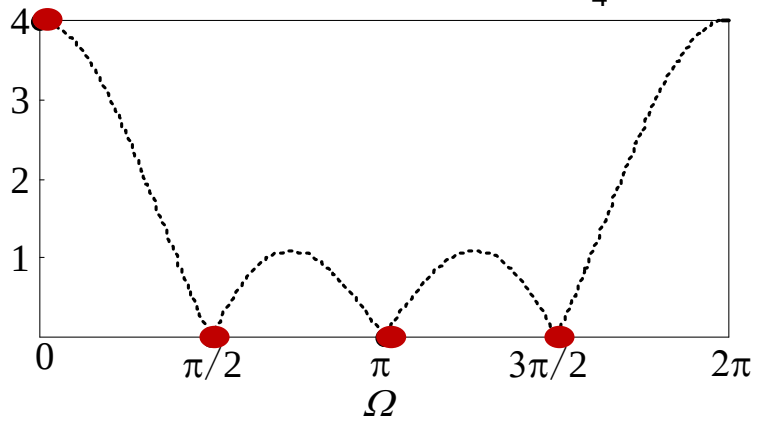
$$x_0[k] = \{1,1,1,1,0,0,0,0\}$$

$$X_0[m] = X(e^{j\Omega})\big|_{\Omega=\frac{2\pi}{8}\cdot m} \qquad \Omega = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$$

Discrete Fourier Transform

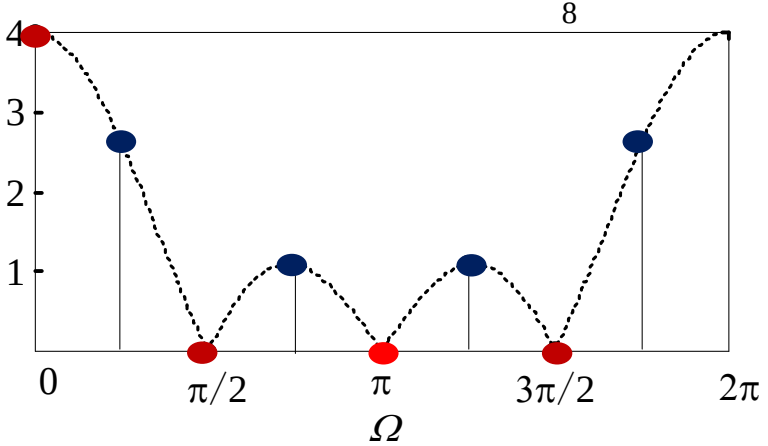
$$x[k]=\{1,1,1,1\}$$

$$X[m]=X(\mathrm{e}^{\mathrm{j}\Omega})\big|_{\Omega=\frac{2\pi}{4}\cdot m}$$



$$x_0[k]=\{1,1,1,1,0,0,0,0\}$$

$$X_0[m]=X(\mathrm{e}^{\mathrm{j}\Omega})\big|_{\Omega=\frac{2\pi}{8}\cdot m}$$



Appending zeros to a sequence allows for more sampling values, enabling the observation of more details within $X(\mathrm{e}^{\mathrm{j}\Omega})$.

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.